

curriculum vitae

Computational Materials Engineer with a Ph.D. in Computational & Experimental Mechanics specializing in physics-based surrogate modeling, multi-scale simulation, and experimental-numerical methods. Proven ability to bridge high-fidelity physics simulations with practical engineering predictions through large-scale data curation, custom Python tooling, and production-grade surrogate models. Strong background in data-driven reduced-order modeling, uncertainty quantification, and deploying scalable multi-physics pipelines in HPC environments. Passionate about using physics-informed and data-driven methods to democratize advanced simulation and enable digital twin applications.

Personalia

Surname:	Ruybalid
Given names	Andre, Paul
Degrees:	Ph.D. MSc. BSc.
City:	Los Alamos, New Mexico, U.S.A.
Phone number:	+1 719 210 1746 (personal mobile)
E-mail:	andreruybalid@gmail.com
Date of birth:	26 October 1987
Place of Birth:	Farmington, New Mexico, U.S.A.
Nationality:	American/Dutch

Professional experience

2023 - Present

R&D Engineer, Los Alamos National Laboratory

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

- ❖ Architected and deployed “LAROMance”, a physics-informed surrogate modeling framework for viscoplastic creep in nuclear reactor materials, enabling 1000x+ speedup over full-order simulations while preserving mechanistic accuracy. Two-time R&D 100 Award Finalist.
- ❖ Designed large-scale simulation pipelines that curated datasets from tens of thousands of low-length-scale simulations, applying ablation studies and dimensionality reduction to optimize model fidelity vs. computational cost.
- ❖ Developed and integrated surrogate constitutive models into the MOOSE/BISON finite element framework, including Verification & Validation (V&V), and large-scale uncertainty quantification (UQ).
- ❖ Built modular Python tooling for guided workflow orchestration, data processing, and visualization, accelerating analysis cycles and enabling reproducible, production-grade deployments of physics-based surrogates.

2021 – 2023

Postdoctoral Researcher, Los Alamos National Laboratory

Materials Science and Technology Division (MST-8) | Los Alamos, NM, U.S.A.

- ❖ Developed a novel surrogate modeling methodology specifically for history-dependent material behavior, using regression methods on data generated from spectral (FFT) polycrystal simulations.
- ❖ Curated and managed large-scale simulation datasets from thousands of high-fidelity runs; designed and implemented preprocessing, feature engineering, and surrogate model calibration pipelines.

2019 – 2020

Design Engineer, ASML

DUV Def & Metrology group of Defectivity Department | Eindhoven, The Netherlands

Developed and maintained complex numerical algorithms and custom MATLAB tooling for high-precision nanoscale defect detection and geometry analysis on DUV lithography reticles. Designed, validated, and optimized metrology and data processing pipelines through large-scale data analysis and cleanroom experimentation, delivering production-ready solutions under tight deadlines in cross-functional industrial environments.

2013 – 2019

PhD Researcher, Eindhoven University of Technology

*Computational and Experimental Mechanics | Prof. Dr. Marc Geers & Dr. Johan Hoefnagels.
Philips Research, Materials Innovation Institute | Eindhoven, The Netherlands*

- ❖ Designed and implemented an integrated experimental-numerical framework combining Finite Element Modeling in MSC Marc, image processing, and Digital Image Correlation (IDIC) for full-field material characterization and inverse parameter identification.
- ❖ Developed virtual experimentation frameworks by applying known displacement fields to deform experimental images, then tested new Digital Image Correlation (IDIC) algorithms through inverse identification and sensitivity analysis.

Education

*Eindhoven University of Technology, the Netherlands
Department of Mechanical Engineering
Division: Computational and Experimental Mechanics
Research Group: Mechanics of Materials*

2013 – 2019

PhD. in Mechanical Engineering

Title: “Identification of multi-layer interface systems: a full-field, micro-mechanical approach.”

2013 – 2019

Graduate School of Engineering Mechanics

Advanced Mechanics Courses

2010 – 2013

MSc. Mechanical Engineering

Minor: Biomedical Engineering

2006 – 2010

BSc. Mechanical Engineering

2000 – 2006

Research-oriented secondary education (VWO)

St.-Janscollege, Hoensbroek, The Netherlands

Technical Skills

Programming & Software Tools	Python, MATLAB, C++, Linux/bash scripting, SLURM (HPC), MSC Marc/Mentat, Paraview, LLM-assisted development (Continue.dev in VS Code)
Scientific Computing	Physics-based surrogate modeling, reduced-order modeling, spectral methods (FFT), finite element method (FEM), sparse linear solvers and regression, large-scale dataset curation & ablation studies, inverse optimization, uncertainty quantification, feature engineering
Software Development	Custom CAE tool & pipeline development, workflow automation, interactive Python tooling (GUI), Python-to-C++ integration, production-grade code development, modular design, Git
Domain Expertise	Computational mechanics, multi-scale & multi-physics modeling, continuum mechanics, viscoplasticity and creep modeling, cohesive zone and interface mechanics, plasticity and dislocation mechanics
Technical Approaches	In-situ small-scale mechanical testing, high-resolution microscopy (SEM, SPM, optical profilometry), Digital Image Correlation (IDIC), experimental-numerical frameworks

Selected journal publications (peer-reviewed)

P. Robbe, A.P. Ruybalid, A. Hegde, C. Bonneville, H.N. Najm, L. Capolungo and C. Safta . A comparison of surrogate constitutive models for viscoplastic creep simulation of HT-9 steel. *Computational Materials Science*, 262, 114367 (2026). <https://doi.org/10.1016/j.commatsci.2025.114367>.

A.P. Ruybalid, A. Tallman, W. Wen, C. Matthews, L. Capolungo. Data-Driven Surrogate Modeling with Microstructure-Sensitivity of Viscoplastic Creep in Grade 91 Steel. *Integrating Materials and Manufacturing Innovation* 13, 895–914 (2024). <https://doi.org/10.1007/s40192-024-00377-z>

A.P. Ruybalid, O. van der Sluis, M.G.D. Geers and J.P.M. Hoefnagels. Full-field identification of mixed-mode adhesion properties in a flexible, multi-layer microelectronic material system. *Engineering Fracture Mechanics*, 226, pp. 106879 (2020), <https://doi.org/10.1016/j.engfracmech.2020.106879>.

A.P. Ruybalid, J.P.M. Hoefnagels, O. van der Sluis, M.P.F.H.L. van Maris and M.G.D. Geers. Mixed-mode cohesive zone parameters from integrated digital image correlation on micrographs only. *International Journal of Solids and Structures*, 156-157, pp. 179-193 (2019), <https://doi.org/10.1016/j.ijsolstr.2018.08.010>.